MILITARY SPECIFICATION
CARTRIDGE, CALIBER .50, TRACER, M1

1. SCOPE

1.1 This specification covers Cartridge, Caliber .50, Tracer, M1 for use in caliber .50 weapons.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-636	- Visual Inspection Standards for Small Arms Ammunition through Caliber .50
MIL-STD-644	- Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition

DRAWINGS

U.S. Army Munitions Command

D5544843	- Cartridge, Caliber .50, Tracer, M1
C7643674	- Classification of Cartridge Case Defects
D7553544	- Packing and Marking, Cartridges, Caliber .50; Bulk; Box Ammunition, M2A1, Box Wirebound.
D7553346	- Packing and Marking, Cartridges, Caliber .50, Linked; Box, Ammunition M2A1, Box Wirebound
F8596993	- Packing and Marking, Cartridges, Caliber .50, Linked, M15 or M15A1; Box Ammunition M2A1, Wirebound
IEL-5544843	- Inspection Equipment List for Cartridge, Caliber .50, Tracer, M1

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2.1 (Cont'd)

PUBLICATIONS

AMCR 715-505, Volume 3	- Ammunition Ballistic Acceptance Test Methods - Test Procedures for 7.62mm Cartridges
TECP 700-700, Vol. III	- Manual of Test Methods for Small Arms Ammunition

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- The cartridge shall comply with Drawing D5544843, referenced specifications and the following:

3.2 Bullet extraction.- The force required to extract the bullet from the cartridge case shall be not less than 200 pounds.

3.3 Airtightness of base closure seal.- The bullet shall not release more than one bubble of air under the application of a pressure differential of 2 pounds per square inch (psi) for 5 seconds.

3.4 Residual stress.- The cartridge case shall not split when subjected to a one percent mercurous nitrate solution for 15 minutes.

3.5 Waterproof.- The cartridge shall not release more than one bubble of air when subjected to a pressure differential of $7\frac{1}{2}$ psi for 30 seconds.

3.6 Accuracy.- The average of the mean radii of all targets of the sample cartridges, fired at 600 yards, shall not be greater than 20 inches.

3.7 Action time.- The action time (overall primer ignition, propellant burning and bullet-barrel time) of the cartridge shall not exceed 4 milliseconds.

3.8 Velocity.- The average velocity of the sample cartridges conditioned at 68° to 72° Fahrenheit (F), shall be 2860 feet per second (ft/sec) plus or minus 40 ft/sec for gilding metal clad steel jacketed bullets or 2700 ft/sec plus or minus 40 ft/sec for gilding metal jacketed bullets, at 78 feet from the muzzle of the weapon. The standard deviation of the velocities shall not exceed 60 ft/sec.

3.9 Chamber pressure.- The average chamber pressure of the sample cartridges conditioned at 68° to 72°F shall not exceed 52,000 psi.

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3.10 Trace.- When viewed at night from a line parallel to the plane of the trajectory at each point of observation, the bullet of the tracer cartridge shall exhibit a visual trace of full luminosity from a point not greater than 250 yards from the muzzle of the weapon to a point not less than 1,600 yards from the muzzle.

3.11 Function and casualty.- The cartridge shall function without casualty.

3.12 Workmanship.- The requirements for workmanship are as specified on the applicable drawing, reference specifications and the following:

3.12.1 Metal defects.- The cartridge shall be free of folds, wrinkles, deep draw scratches, scaly metal, dents, and other defects.

3.12.2 Foreign matter.- The cartridge shall be free of corrosion, stains, discolorations, dirt, oil and smears of lacquer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 2000 cartridges. The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a government laboratory or such other facility specified in the contract.

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4.2.1.2 Initial production sample failure.- Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Lot identification.- Each lot of ammunition shall be identified as to type, caliber and model, as well as with a lot number and the supplier's identification as assigned by the procuring activity. Each lot shall be further identified by a Federal Stock Number assigned by the procuring activity.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a class basis in accordance with the classification of defects, Table I, using applicable sampling plans and acceptance criteria of MIL-STD-105. The AQL for Major Class shall be 0.25 percent and the AQL for the Minor Class shall be 1.50 percent.

4.3.2.1 Classification of defects.- The classification of defects shall be as specified in Table I.

TABLE I

No.	Defect and Method of Inspection	Critical	Major	Minor	Major or Minor
	Visual 1/				
	Cartridge				
1	Discolored, dirty, oily, smeared			X	
2	Corroded, or stained, if etched		X		
	Case				
4	Round head		X		
5	Dent				X
6	Split case				
	in K, L or M location	X			
	in I, S or J location		X		
7	Perforated case	X			
8	Draw scratch				X
9	Scratch			X	

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4.3.2.1 (Cont'd)

TABLE I (Cont'd)

No.	Defect and Method of Inspection	Critical	Major	Minor	Major or Minor
10	Beveled underside of head		X		
11	Case mouth not crimped in canneulure		X		
12	Scaly metal				X
13	No chamfer on head (rim)		X		
14	Fold			X	
15	Wrinkle			X	
16	Buckle			X	
17	Bulge			X	
18	Illegible or missing head stamp			X	
19	Defective head			X	
20	Defective mouth			X	
21	No visible evidence of mouth anneal		X		
	Bullet				
22	Dent			X	
23	Scratch			X	
24	Split bullet jacket		X		
25	Loose bullet		X		
26	Missing cannelure		X		
27	Scaly metal				X
28	Upset (crooked) point			X	
29	Exposed steel (clad pocket)			X	
30	Blunt point			X	
31	Defective cannelure			X	
	Primer				
32	No primer	X			
33	Cocked primer	X			
34	Inverted primer	X			
35	Loose primer		X		
36	Nicked or dented primer			X	
37	No waterproofing material (Primer pocket joint)			X	
38	Defective crimp			X	
	Gaging				
39	Total length		X		
40	Cartridge profile failure (requiring more than 80 lbs. dead weight to insert in profile and alignment gage)		X		
41	Diameter of extractor groove, max.		X		
42	Diameter of extractor groove, min.			X	
43	Diameter of head		X		
44	Thickness of head		X		

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4.3.2.1 (Cont'd)

TABLE I (Cont'd)

No.	Defect and Method of Inspection	Critical	Major	Minor	Major or Minor
45	Length to shoulder datum		X		
46	Depth of primer		X		
	Weighing				
47	Weight, min 2/	X			

1/ Refer to MIL-STD-636 for visual defects standards for defects 1 through 38.

2/ Each lightweight cartridge shall be disassembled and the propellant weighed. Each such cartridge found to contain 100 grains or more of propellant shall be classed as a major defect. Any cartridge containing less than 100 grains of propellant shall be classed as a critical defect.

4.3.3 Tests.- The tests listed in Table II shall be conducted in accordance with the methods and procedures specified in 4.4.

4.3.3.1 Test sample.- The quantities for the various tests shall be as specified in Table II. Only cartridges having met the visual, dimensional and weight requirements shall be used in the ballistic tests and shall have been selected in such a manner that the sample is representative of the entire lot. The cartridges shall be thoroughly mixed before being divided into samples for the various tests.

TABLE II

Test	Number of Cartridges	Requirement Paragraph
Bullet Extraction 1/	25	3.2
Airtightness of Base Closure Seal 2/	--	3.3
Residual stress (Mercurous Nitrate) 1/	50	3.4
Waterproof 3/	20	3.5
Accuracy 4/	90	3.6
Action Time 1/	50	3.7
Velocity 4/	20	3.8
Chamber pressure 4/	20	3.9
Trace 5/	100	3.10
Function and Casualty 6/	--	3.11
Gun, Machine Cal. 50 Browning M2, HB, (Turret Type)	400	--

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TABLE II (Cont'd)

Test	Number of Cartridges	Requirement Paragraph
Gun, Machine Cal. 50 Browning M2, HB (Flexible Type)	400	--
Gun, Machine Cal. .50, M85 (Fixed Type)	400	--

1/ Failure of two or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If one cartridge fails in the first test, a second sample consisting of double the number of cartridges in the first sample shall be tested. If any failing cartridges are found in the second sample, the lot shall be rejected.

2/ The bullets, extracted from the cartridges in the bullet extraction test, shall be used for the first test. Failure of seven or more bullets to comply with the applicable requirements shall be cause for rejection of the lot. If more than three but less than seven bullets fail in the first test, a second sample consisting of double the number of bullets in the first sample shall be tested. The lot shall be rejected if, in the combined first and second sample, seven or more bullets fail to comply with the applicable requirement.

3/ Failure of nine or more cartridges to comply with the applicable requirements shall be cause for rejection of the lot. If more than four, but less than nine cartridges fail in the first test, a second sample consisting of double the number of cartridges in the first sample shall be tested. The lot shall be rejected if in the combined first and second sample, nine or more cartridges fail to comply with the applicable requirement.

4/ Failure of the cartridges to comply with the applicable requirements shall be cause for rejection of the lot, subject to testing a second sample consisting of double the quantity of cartridges used in the first test. Failure of the cartridges in the second sample to comply with the applicable requirement shall be cause for rejection of the lot.

5/ Failure of thirty-one of more cartridges to comply with the applicable requirements shall be cause for rejection of the lot. If more than ten but less than thirty-one cartridges fail in the first test, a second sample consisting of double the number of cartridges in the first sample shall be tested. The lot shall be rejected if in the combined first and second sample, thirty-one or more cartridges fail to comply with the applicable requirement.

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6/ The lot shall be rejected when function and casualty defects plus firing defects observed in all other firing tests exceed the acceptance number for the cumulative sample in Table III. If the number of defects found in the first test exceeds the acceptance number for the first sample, but is equal to or less than the acceptance number for the cumulative sample, a second sample, consisting of double the quantities specified under function and casualty test, shall be fired in all the service weapons specified therefor. This procedure shall apply regardless of the weapon of weapons in which the firing defects occurred in the first test. If the total number of defects in the combined first and second sample exceeds the acceptance number for the cumulative sample, the lot shall be rejected. If, in testing a second sample, defects other than those for which the second sample is being tested should occur to the extent that they exceed the acceptance number for the cumulative sample, the lot shall be rejected.

4.3.3.2 Firing defects.- Firing defects and acceptance numbers shall be as specified in Table III.

TABLE III

Defects	ACCEPTANCE First Sample	ACCEPTANCE Cumulative (1st & Second Sample)
1. Misfire	1	2
2. Bullet remaining in bore 1/	0	
3. Primer leak		
a. Perforation in firing pin indent in primer cup	24	60
b. Escape of gas through primer cup other than 3a.	7	15
c. Escape of gas around primer cup more than 50% of periphery	15	31
d. Blown primer or primer falls out of pocket on retraction of bolt	0	1
4. Case casualties		
a. Longitudinal split 2/		
(1) Neck and shoulder (I or S)	29	67
(2) Body (J)	7	15
(3) Body (K)	1	2
(4) To head (L)	0	1
(5) Through head (M)	0	1
b. Circumferential rupture 2/		
(1) Partial, shoulder or body (J & S)	2	4
(2) Partial, body (K)	0	1
(3) Partial, head (L)	0	1
(4) Complete	0	1
5. Failure to extract	0	1
6. Weapon stoppage 3/	0	1

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4.3.3.2 (Cont'd)

1/ No second sample permitted. Lot shall be rejected.

2/ For location of defects indicated by letters in parentheses, see Drawing C7643674.

3/ All stoppages attributable to the ammunition, with the exception of misfire, complete rupture or failure to extract, observed in all tests shall be included.

4.3.4 Packaging, packing and marking inspection.- During or immediately prior to the packaging operation, 100 percent examination of the cartridges shall be performed to ascertain that the cartridge type conforms to the drawing. Occurrence of a high pressure test, dummy or blank cartridge shall be classed as a critical defect. Occurrence of any incorrect type other than those listed shall be classed as a major defect. All non-conforming cartridges shall be rejected. Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644 as applicable to the drawing.

4.3.5 Inspection equipment.- The examination and tests shall be made using the equipment listed on IEL-5544843.

4.4 Test methods and procedures.

4.4.1 Bullet extraction.- The cartridges shall be tested in an approved bullet extractor machine. The rate of travel of the test head shall be not less than three or more than six inches per minute.

4.4.2 Airtightness of base closure seal.- The bullets shall be placed base up under approximately 1.5 inch head of freshly boiled water in a vacuum desiccator or vacuum jar test unit. The desiccator or jar shall be evacuated to a pressure of 2 pounds per square inch (4 inches of mercury) below atmospheric pressure and held at that pressure for not less than 5 seconds. The number of bubbles liberated from each bullet shall be observed.

4.4.3 Residual stress (Mercurous Nitrate).- The test shall be conducted in accordance with TECP 700-700, Vol. III.

4.4.4 Waterproof.- The ammunition to be tested, not to exceed five cartridges at one time, shall be placed horizontally on a perforated tray. The tray shall be placed in a desiccatory containing sufficient quantity of freshly boiled water to maintain a head of 2 to $2\frac{1}{2}$ inches. The desiccator shall be evacuated to a pressure of $7\frac{1}{2}$ pounds per square inch (15 inches of mercury) below atmospheric pressure and held at that pressure for 30 seconds. The number of bubbles liberated from the mouth or primer of each cartridge shall be observed.

4.4.5 Accuracy.- The test shall be conducted in accordance with TECP 700-700, Vol. III.

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4.4.6 Action time.- The test shall be conducted in accordance with methods and procedures of AMCR 715-505, Volume 3 using equipment listed in IEL-5544843.

4.4.7 Velocity.- The test shall be conducted in accordance with TECP 700-700, Vol. III.

4.4.8 Chamber pressure.- The test shall be conducted in accordance with TECP 700-700, Vol. III.

4.4.9 Trace.- The test shall be conducted in accordance with methods and procedures of TECP 700-700, Vol. III and the following: Observation for trace performance shall be conducted on a range of at least 1600 yards in length. All trace testing shall be performed at night. Observation for trace performance shall be made at the weapon and at 250 yards and 1600 yards beyond the muzzle of the gun on a line parallel to and approximately 75 yards from the plane of trajectory. Lights or other suitable visible markers shall be placed opposite the 250 and 1600 yards observing points to insure a line of sight from each point perpendicular to the plane of the trajectory. The ammunition shall not be conditioned to temperature but shall be fired under existing conditions. Three cartridges shall be fired to sight, warm and foul each weapon prior to the test. The sample cartridges shall be fired in regular sequence permitting sufficient time between shots for each observer to record trace characteristics at each observation point. Each cartridge shall be considered as a defective once only in the event it fails at more than one observation point. During the performance of this test, observations shall also be made for (a) bullet bursting before striking terminal or target, (b) trace muzzle flash and (c) erratic flight.

4.4.10 Function and casualty.- In these firings, the weapons shall be at room temperature at the beginning of the test, and the machine guns shall be cooled between bursts. The number of cartridges to be fired shall be as specified in Table II. The test shall be conducted in accordance with TECP 700-700, Vol. III and as indicated below:

a. Gun, Machine, Caliber .50 Browning, M2, Heavy Barrel (Turret Type) - Fire in bursts of 100 cartridges.

b. Gun, Machine, Caliber .50 Browning M2, Heavy Barrel (Flexible Type) - Fire in bursts of 100 cartridges.

c. Gun, Machine, Caliber .50, Tank M85 (Fixed) - Fire in bursts of 100 cartridges.

4.4.11 Defect penalty.- In any ballistic test, except function and casualty, in which the occurrence of a firing defect listed in Table III prevents the obtaining of a reliable result for the characteristic being tested, an additional shot shall be fired. That particular test shall not be penalized, but the total ballistic sample shall be penalized for such defects in accordance with Table III.

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5. PREPARATION FOR DELIVERY

5.1 Packing, Level A (Worldwide shipment).- The cartridge shall be packed in accordance with Drawing D7553346, D7553544 or F8596993.

5.2 Marking and labeling.- Packing boxes shall be marked and labeled in accordance with the applicable drawing cited in 5.1.

6. NOTES

6.1 Ordering data.- Invitations for bids and contracts or orders will specify the following:

6.1.1 Title, number and date of this specification.

6.1.2 Type and level of packing.

6.1.3 Provisions for the supply, maintenance and disposition of mandatory ballistic test equipment for acceptance inspection purposes.

6.1.4 Provision for the submission of acceptance inspection reports containing final inspection results for each lot of ammunition presented to the Government.

6.1.5 Requirements for contractor to provide and maintain an inspection system in accordance with MIL-I-45208, Inspection System Requirements.

Custodian:

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Preparing activity:

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Project No. 1305-A501

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